



National
Semiconductor™

Discrete POWER & Signal
Technologies

NPN General Purpose Amplifiers and Switches

Device No.	Case Style	V _{CEO} (V) Min	V _{CE0} (V) Min	V _{EB0} (V) Min	I _{CBO} V _{CB} (nA) @ (V) Max	h _{FE} @ I _C & V _{CE} Min Max (mA) (V)	V _{CE(SAT)} V _{BE(SAT)} (V) & (V) @ Max Min Max	I _C (mA) Max	C _{ob} (pF) Max	f _T (MHz) @ Min	I _C (mA) Max	t _{on} (ns) Max	NF (dB) Max	Test Conditions	Process No.
2N3390	TO-92 (94)	25	25	5	100 18	400 800 2 4.5			10						10
2N3391A	TO-92 (94)	25	25	5	100 18	250 500 2 4.5			10				5	(Note 5)	10
2N3392	TO-92 (94)	25	25	5	100 18	150 300 2 4.5			10						10
2N3393	TO-92 (94)	25	25	5	100 18	90 180 2 4.5			10						10
2N3415	TO-92 (94)	25	25	5	100 25	180 540 2 4.5	0.3 0.6 1.3 50								10
2N3416	TO-92 (94)	50	50	5	100 25	75 225 2 4.5	0.3 0.6 1.3 50								10
2N3417	TO-92 (94)	50	50	5	100 25	180 540 2 4.5	0.3 0.6 1.3 50								10
2N3704	TO-92 (94)	50	30	5	100 20	100 300 50 2	0.6 100		12	100 50					10
2N5172	TO-92 (94)	25	25	5	100 25	100 500 10 10	0.25 10		10						10
MPS8088	TO-92 (92)	60	60	6	100 60	100 300 1 5 100 10 5 75 100 5	0.3 100		6	150 10					10
MPSA20	TO-92 (92)		40	4	100 30	40 400 5 10			4	125 5					10